Checkpoint 6 Science — Practice Test 2

Paper 1: 35 minutes · calculator allowed · Paper 2: 35 minutes · Total 80 marks

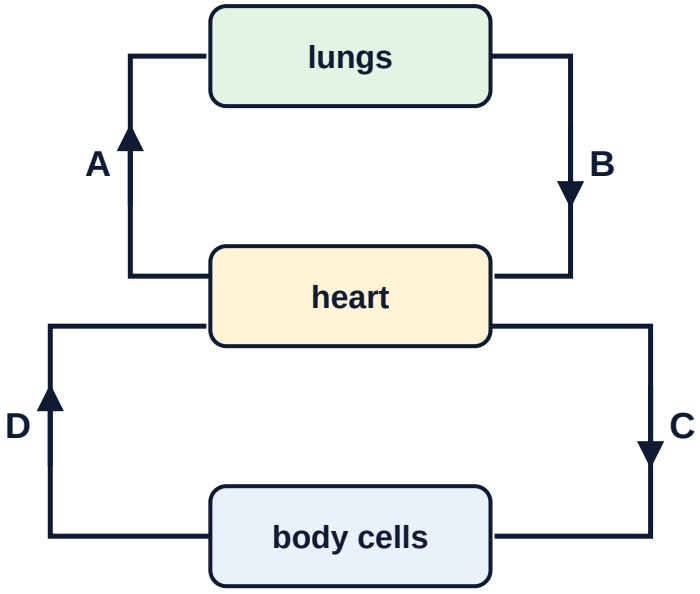
Answer all questions. Write your answers in the spaces provided. Show your working where there is space for it. The number of marks for each question is shown in brackets [ ].

Name _____ Class _____ Date _____ Mark: ____ / 80

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 35 minutes — 40 marks

1 Karim plays in a school football match. The diagram is a simple model of how blood travels around his body. The arrows show the direction in which the blood flows. A, B, C and D are blood vessels. **[1]**



(a) Which letter shows the blood vessel that carries blood **from the heart to the body cells**?

(b) Blood in vessel B has just come from the lungs. Name the gas that the blood has picked up in the lungs. **[1]**

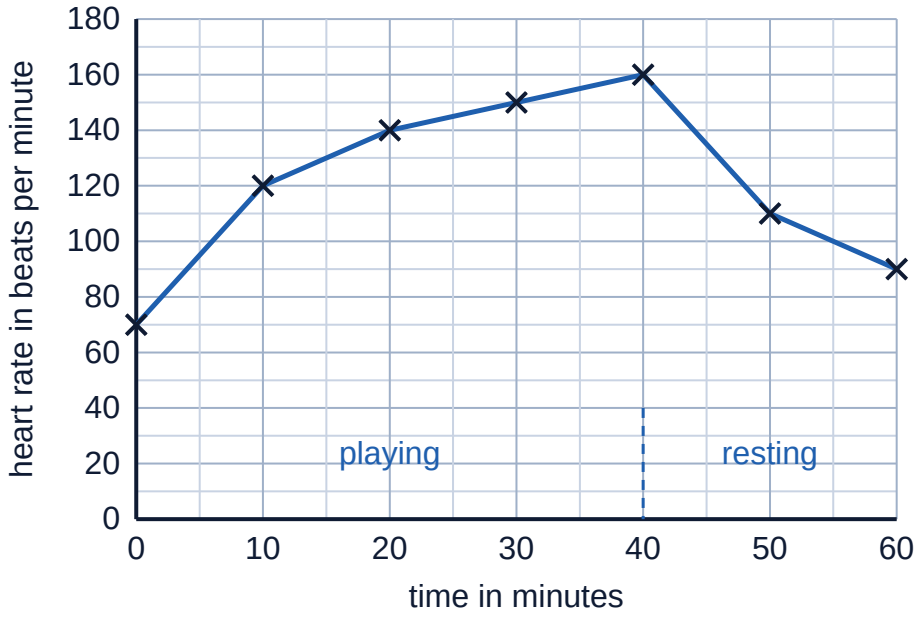
(c) In Karim's leg muscles, blood flows through very narrow blood vessels with walls only one cell thick. Name these blood vessels. **[1]**

(d) Which sentence describes the job of Karim's circulatory system? **[1]**

- ☐ A) It breaks down his food into small pieces.
- ☐ B) It moves the bones in his legs when he runs.
- ☐ C) It carries oxygen, nutrients and waste around his body.
- ☐ D) It takes air into his body and lets air out.

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2 Karim wears a heart-rate watch. He plays football for the first 40 minutes, then sits and rests on the bench from 40 to 60 minutes. The graph shows his heart rate. **[1]**



(a) What was Karim's heart rate after 20 minutes?

Working

Answer ____ beats per minute

(b) Work out how much Karim's heart rate increased from the start (0 minutes) to 40 minutes. **[1]**

Working

Answer ____ beats per minute

(c) Explain why Karim's heart rate increased while he was playing football. **[2]**

3 Mum is cooking dinner in the family kitchen. She cuts up raw chicken and then makes a salad. **[1]**

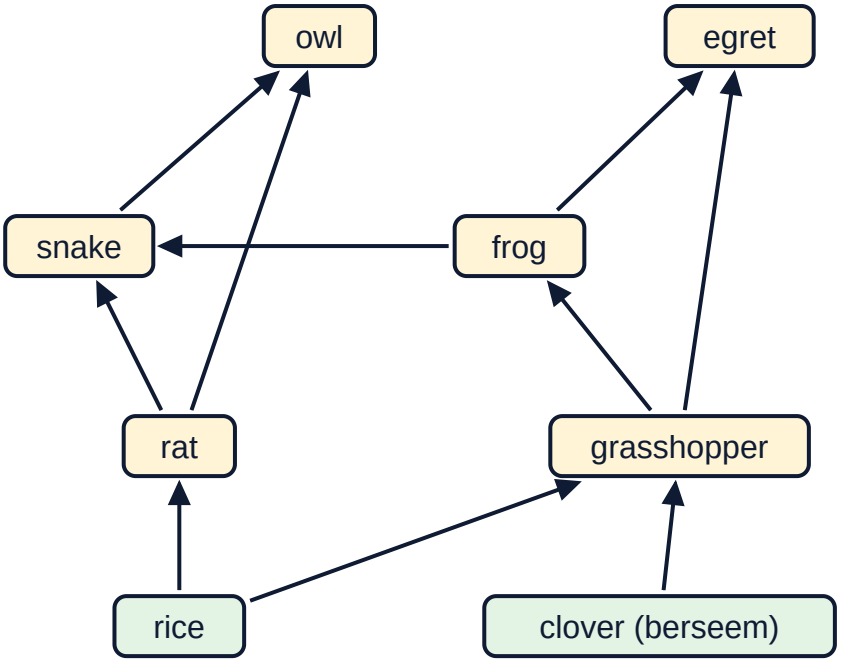
(a) Raw chicken can carry harmful bacteria. Which action is best for stopping these bacteria from getting into the salad?

- ☐ A) Use the same knife for the chicken and the salad, but cut the salad quickly.
- ☐ B) Leave the salad on the worktop near the chicken for a few hours.
- ☐ C) Rinse the chicken in the sink next to the salad bowl.
- ☐ D) Wash the board, the knife and her hands with hot soapy water after cutting the chicken.

(b) If a few bacteria are swallowed with food, most of them are killed in the stomach. Name the substance in the stomach that kills them. **[1]**

(c) In summer the family leaves the kitchen window open at night and mosquitoes fly in. Some mosquitoes can carry pathogens from one person to another. Give **two** ways the family can avoid being bitten by mosquitoes. **[2]**

4 The diagram shows part of a food web on a farm in the Nile Delta. The farmer grows rice and clover (berseem). **[1]**



(a) Name **one** producer in this food web.

(b) How many food chains in this food web include the **rat**? **[1]**

Working

Answer

(c) A disease kills most of the frogs. What is most likely to happen to the number of grasshoppers at first? **[1]**

- ☐ A) It will decrease, because grasshoppers eat frogs.
- ☐ B) It will increase, because fewer grasshoppers are eaten.
- ☐ C) It will stay the same, because grasshoppers eat plants.
- ☐ D) All the grasshoppers will die.

(d) The farmer puts out rat poison. The poison is toxic and stays inside the bodies of animals that take it in. A few weeks later, an owl on the farm is found dead. Explain how the poison got into the owl's body. **[2]**

(e) Name the original source of the energy for every food chain in this food web. **[1]**

5 The table shows the melting points and boiling points of some substances found in the family kitchen. **[1]**

Substance	Melting point in °C	Boiling point in °C
water	0	100
salt	801	1413
ethanol (cleaning spirit)	−114	78
pure ethanoic acid (used to make vinegar)	17	118

(a) What is the state of salt at room temperature (25 °C)? Write solid, liquid or gas.

(b) Which substance in the table is a **gas** at 90 °C? **[1]**

(c) On a cold winter morning the kitchen is at 10 °C. What is the state of the pure ethanoic acid in its bottle? **[1]**

6 Salma mixes lemon juice with baking soda in a glass in the kitchen. The mixture fizzes. She puts a thermometer in the mixture. **[1]**

temperature at the start = 22 °C

temperature after 1 minute = 17 °C

The word equation for the reaction is:

citric acid + sodium hydrogencarbonate → sodium citrate + water + carbon dioxide

(a) By how many degrees did the temperature fall?

Working

Answer ☐ °C

(b) Give **one other** observation that shows a chemical reaction is happening. **[1]**

(c) How many **reactants** are there in this reaction? **[1]**

Working

Answer ☐

(d) Salma also cooks dinner. Which of these changes is **reversible**? **[1]**

- ☐ A) melting butter in a warm pan
- ☐ B) frying an egg
- ☐ C) toasting bread
- ☐ D) baking a cake

7 Youssef investigates friction on the playground slide. He sticks a different material to the bottom of the same wooden block, lets the block go from the top of the slide and measures the time it takes to reach the bottom. He does each test three times. **[1]**

Material on the block	Test 1 time in s	Test 2 time in s	Test 3 time in s	Mean time in s
plastic	2.1	2.3	2.2	2.2
cotton cloth	2.8	2.9	3.0	2.9
carpet	4.6	3.5	4.8	?
sandpaper	6.2	6.0	6.4	6.2

(a) Which is the **independent** variable in Youssef's investigation (the variable he changes)?

- ☐ A) the time taken to reach the bottom
- ☐ B) the material on the bottom of the block
- ☐ C) the length of the slide
- ☐ D) the mass of the block

(b) Name the **dependent** variable (the variable he measures). **[1]**

(c) Give **one** variable Youssef must keep the same to make this a fair test. **[1]**

(d) One result in the table is anomalous (it does not fit the pattern). Write down this time. **[1]**

Working

Answer ____ s

(e) Work out the mean time for carpet. Do **not** use the anomalous result. **[1]**

Working

Answer ____ s

(f) Write a conclusion for Youssef's investigation. Use your scientific knowledge to explain it. **[2]**

8 The family kitchen has a granite worktop, a marble floor and a sandstone wall tile. **[1]**

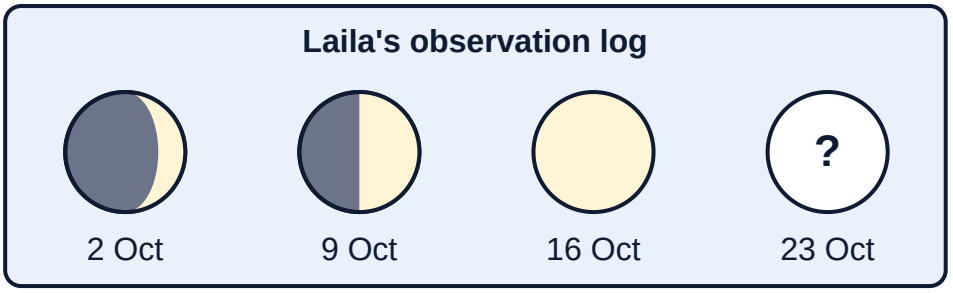
(a) Granite is made of large crystals that fit tightly together. It formed when hot liquid rock cooled slowly underground. What type of rock is granite?

(b) Marble is a metamorphic rock. How did the marble form? **[1]**

- ☐ A) Sand grains were stuck together in layers under the sea.
- ☐ B) Lava cooled quickly on the surface of the Earth.
- ☐ C) Pieces of rock were carried away by a river.
- ☐ D) Limestone was heated and squeezed deep underground, without melting.

(c) The sandstone tile has a fossil of a small sea shell in it. Describe how this fossil formed in sedimentary rock. **[2]**

9 Laila keeps a night sky observation log. Every week she draws the Moon as she sees it from her balcony in Egypt. **[1]**



Possible drawings for 23 October:



(a) Which drawing, A, B, C or D, shows what Laila will most likely see on 23 October?

- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

(b) About one week after 23 October, Laila cannot see the Moon at all. What is this phase of the Moon called? **[1]**

(c) From 2 October to 16 October, the lit part of the Moon that Laila sees gets bigger. Explain why. **[2]**

Paper 2 — 35 minutes — 40 marks

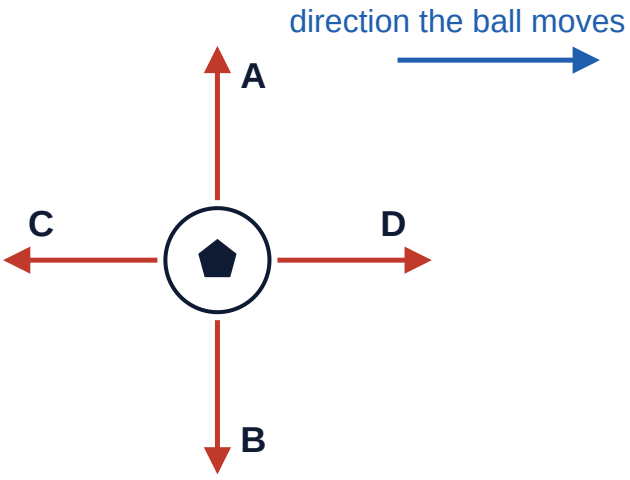
1 This question is about growing up. **[1]**

(a) Which change happens to **both** boys and girls during puberty?

- ☐ A) The voice becomes much deeper.
- ☐ B) Hair grows under the arms.
- ☐ C) Breasts develop.
- ☐ D) The shoulders become much wider.

(b) Name the organs in the female reproductive system that make egg cells. **[1]**

2 Ahmed kicks a football high into the air during a match. The diagram shows the ball at the top of its path, when it is moving to the right. The arrows A, B, C and D are possible forces. **[1]**



(a) Which arrow shows the **weight** of the ball?

(b) Which arrow shows the **air resistance** on the ball? **[1]**

(c) Later, the goalkeeper catches a ball that is moving towards him. Describe the effect of the force from his hands on the ball. **[1]**

(d) Ahmed breathes faster during the match. Which sentence describes what happens in his lungs? **[1]**

- ☐ A) Oxygen moves from his blood into the air in his lungs.
- ☐ B) Carbon dioxide moves from the air in his lungs into his blood.
- ☐ C) Food moves from the air in his lungs into his blood.
- ☐ D) Oxygen moves from the air in his lungs into his blood.

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3 Mona plays in the playground. Her mass is 35 kg. On Earth, gravity is 10 N/kg. **[1]**

(a) Calculate Mona's weight on Earth.

Working

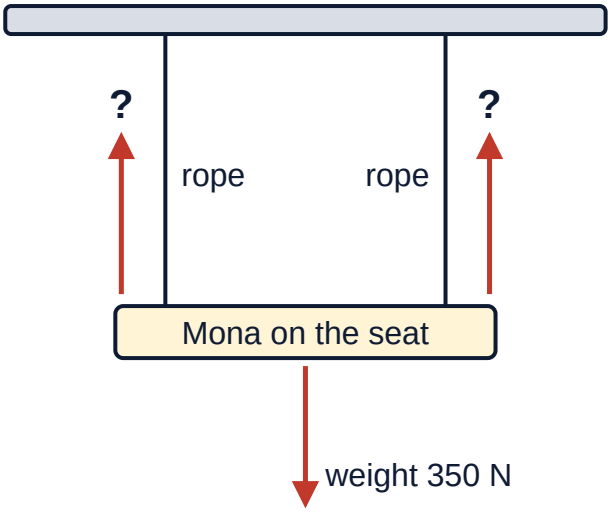
Answer ____ N

(b) Mona imagines playing on the Moon, where gravity is about 1.6 N/kg. Which sentence about Mona on the Moon is correct? **[1]**

- ☐ A) Her mass is still 35 kg, but her weight is less than 350 N.
- ☐ B) Her mass is less than 35 kg, but her weight is still 350 N.
- ☐ C) Her mass and her weight are both less than on Earth.
- ☐ D) Her mass and her weight are both the same as on Earth.

4 Mona plays in the playground. Her mass is 35 kg. On Earth, gravity is 10 N/kg. **[1]**

Mona now sits still on a swing. The diagram shows the forces on Mona and the seat. Ignore the weight of the seat and the ropes.



(c) Mona is not moving. What is the **total** upward force of the two ropes?

Working

Answer ____ N

(d) The two ropes pull up equally. What is the force in **each** rope? **[1]**

Working

Answer ____ N

5 At a model boat competition, Omar's team tests how the mass of cargo affects how deep their boat sits in the water. They measure the depth of the bottom of the boat below the water surface. **[1]**

Mass of cargo in g	**0**	**20**	**40**	**60**	**80**
Depth of boat in cm	1.0	1.5	2.0	?	3.0

(a) Which is the **independent** variable in this investigation?

- ☐ A) the depth of the boat in the water
- ☐ B) the size of the water tank
- ☐ C) the mass of the cargo
- ☐ D) the shape of the boat

(b) Describe the pattern in the results. **[1]**

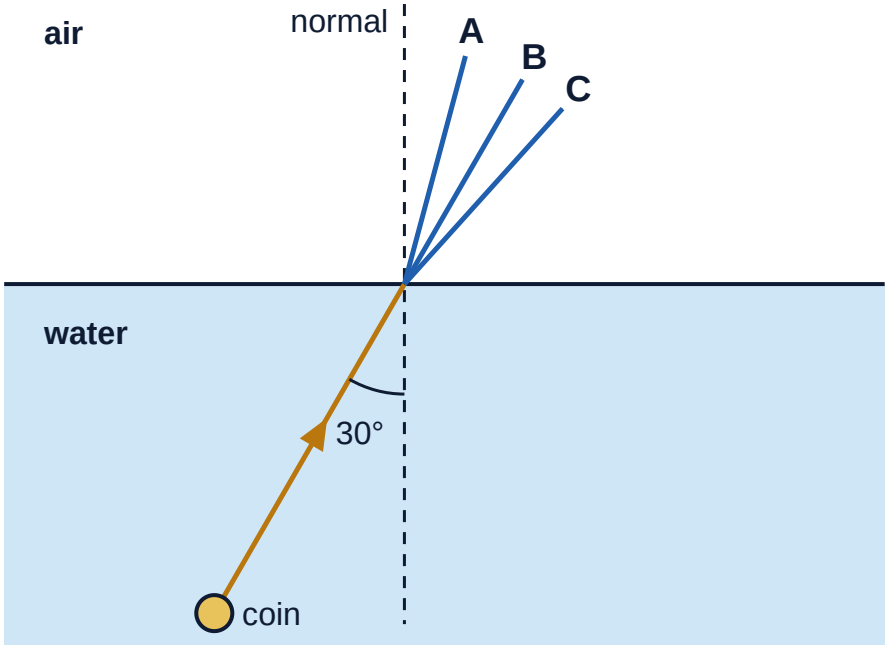
(c) Use the pattern to predict the depth of the boat when it carries 60 g of cargo. **[1]**

Working

Answer ____ cm

(d) Another team rolls the same mass of modelling clay into a solid ball, and it sinks. Omar's team shapes the same mass of clay into a wide, hollow boat, and it floats. Explain why the boat floats but the ball sinks. **[1]**

6 At the model boat competition, a judge drops a coin into the pool. Light from the coin travels up through the water and into the air. Lines A, B and C show three possible paths for the light in the air. **[1]**

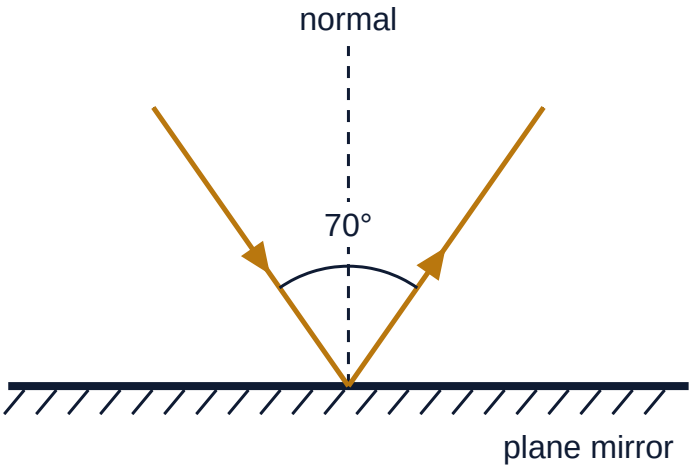


(a) Which line, A, B or C, shows the correct path of the light in the air?

(b) What is the name for the change in direction of light when it passes from one material into another? **[1]**

7 (continued) **[1]**

The judge also uses a plane mirror. The diagram shows a ray of light reflected from the mirror. The angle between the ray going in and the ray coming out is 70°.

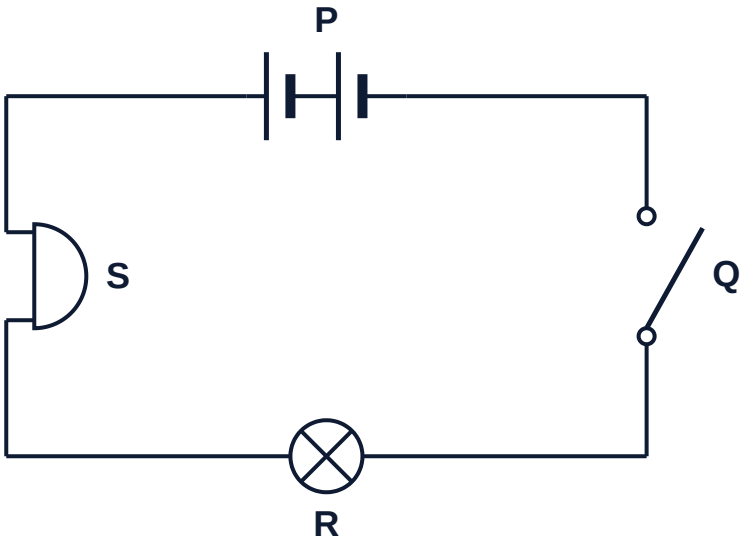


(c) What is the angle of incidence?

Working

Answer °

8 Rana fits a horn and a light to her model boat. The diagram shows her circuit. P, Q, R and S are the components. **[1]**



(a) Which letter labels the **buzzer**?

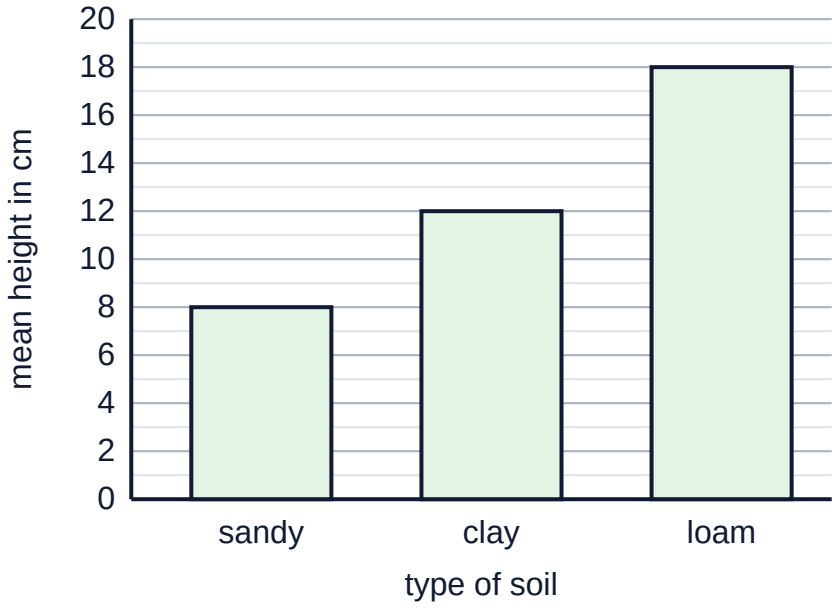
(b) Is switch Q open or closed? Write open or closed. **[1]**

(c) Explain why the buzzer is not sounding. **[1]**

(d) Rana wants to add a second, identical lamp so that, when the switch is closed, **each** lamp is as bright as lamp R would be on its own. How should she connect the new lamp? **[1]**

- ☐ A) in series with lamp R
- ☐ B) in parallel with lamp R
- ☐ C) in place of the battery
- ☐ D) across the open switch

9 Ali lives on a farm in the Nile Delta. He grows bean plants in pots of three different soils from the farm: sandy soil, clay soil and loam (a mixture of sand, clay and humus). He grows four plants in each soil and measures their heights after three weeks. The bar chart shows the mean height for each soil. **[1]**



(a) What is the **dependent** variable in Ali's investigation?

(b) Give **one** variable Ali must keep the same to make his test fair. **[1]**

(c) How much taller, on average, were the plants in loam than the plants in sandy soil? **[1]**

Working

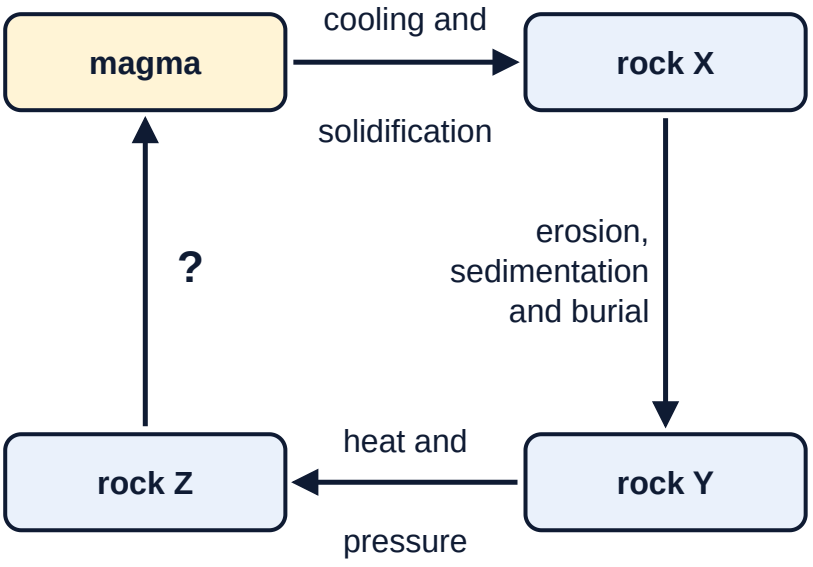
Answer _____ cm

(d) Explain why the plants grew least well in the sandy soil. **[2]**

(e) Ali wants to grow beans in the sandy soil next year. Which change would most help the plants to grow? **[1]**

- ☐ A) Mix in compost (humus) to the sandy soil.
- ☐ B) Add more sand to the soil.
- ☐ C) Remove all the small stones and humus.
- ☐ D) Press the soil down hard so that no air is left.

10 The diagram shows part of the rock cycle. X, Y and Z are three types of rock. **[1]**



(a) Name the type of rock X.

(b) Name the type of rock Z. **[1]**

(c) Name the process shown by the arrow marked ? that changes rock Z back into magma. **[1]**

11 Laila's night sky observation log also records three planets that she saw from Egypt: Venus, Mars and Jupiter. **[1]**

(a) Which sentence correctly describes the movements in our Solar System?

- ☐ A) The Sun and the planets orbit the Earth.
- ☐ B) The Moon orbits the Sun, and the Earth stays still.
- ☐ C) The planets orbit the Moon, and the Moon orbits the Sun.
- ☐ D) The planets orbit the Sun, and the Moon orbits the Earth.

(b) Which of the three planets in Laila's log is **closest** to the Sun? **[1]**

(c) Which planet in Laila's log has its orbit between the orbit of the Earth and the orbit of Jupiter? **[1]**

12 Dad cooks pasta in a pan of salty water on the cooker in the family kitchen. **[1]**

(a) The water in the pan boils. Which sentence describes **boiling** but **not** evaporation?

- ☐ A) The liquid changes into a gas.
- ☐ B) It only happens at the surface of the liquid.
- ☐ C) Bubbles of gas form all through the liquid, at one temperature.
- ☐ D) It can happen at any temperature.

(b) After dinner, Dad wipes the worktop with a wet cloth. One hour later the worktop is dry, even though the water never got hot enough to boil. Name the process that dried the worktop. **[1]**

13 Dad cooks pasta in a pan of salty water on the cooker in the family kitchen. **[1]**
Dad reads that 36 g of salt can dissolve in 100 ml of water at 20 °C, and 39 g can dissolve in 100 ml of water at 100 °C.

(c) How many more grams of salt can dissolve in 100 ml of water at 100 °C than at 20 °C?

Working

Answer ____ g

14 Dad cooks pasta in a pan of salty water on the cooker in the family kitchen. **[2]**

(d) Dad notices that salt dissolves **faster** in hot water than in cold water. Use the particle model to explain why.

15 The family is getting ready for a party in the kitchen. **[1]**

(a) Hana ties two blown-up balloons to the ends of a stick. She hangs the stick from a string so that it balances. Then she pops one balloon. What happens?

- ☐ A) The stick stays balanced, because air has no mass.
- ☐ B) The end with the popped balloon goes down.
- ☐ C) The end with the full balloon goes down, because the air in it has mass.
- ☐ D) Both ends go up.

(b) Hana puts four spoons into cups of hot tea. Which spoon will feel hot fastest when she touches its handle? **[1]**

- ☐ A) a steel spoon
- ☐ B) a wooden spoon
- ☐ C) a plastic spoon
- ☐ D) a bamboo spoon